Generating Reaction Path Hypotheses

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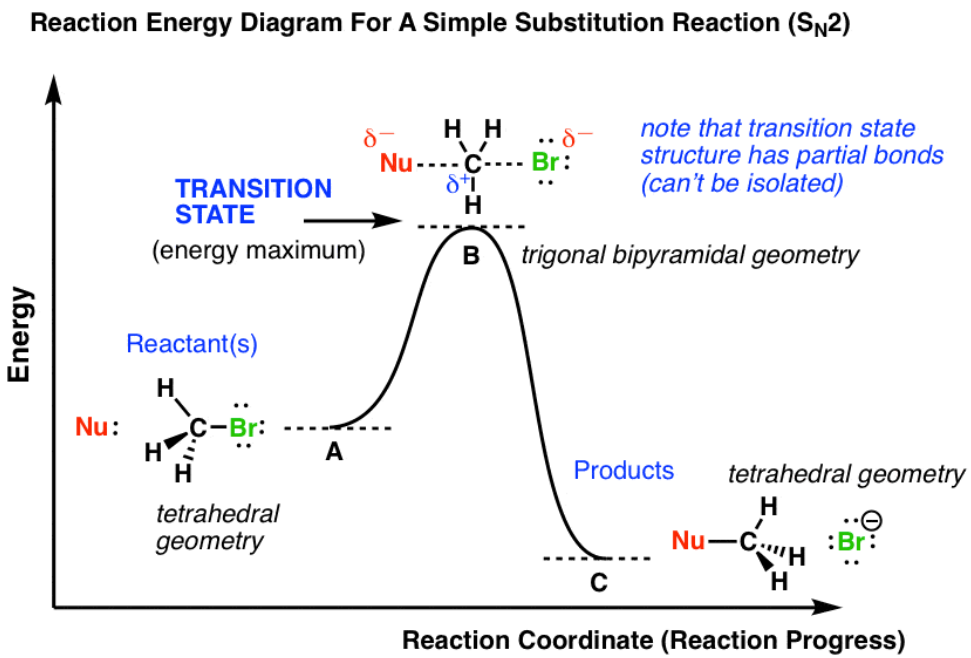
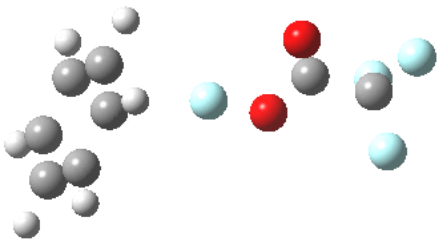
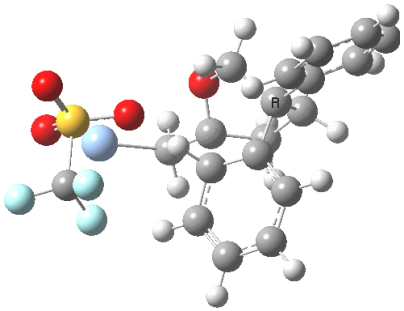
What about reaction pathways?

- Transition states control reaction rates and mechanisms
- Computing TSs is expensive and brittle
- We can reduce cost across reactions



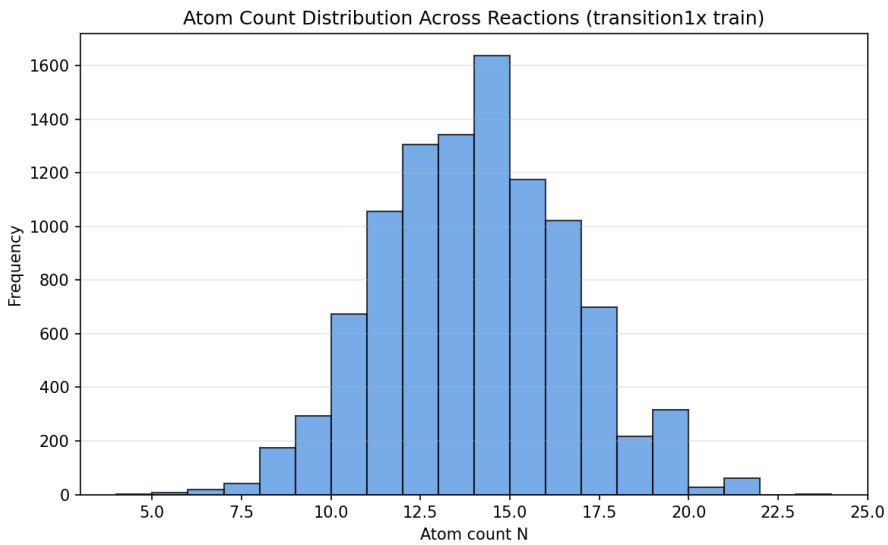




Aromatic electrophilic substitution as part of a catalytic process

Transition1x dataset

- ~10,000 reactions (Transition1x)
- Atom-mapped reactants, products, and transition states
- All geometries provided as Cartesian coordinates
- Energies and forces available (DFT level)
- Multiple TS guesses (midpoint, xTB, NEB-CI, etc.)

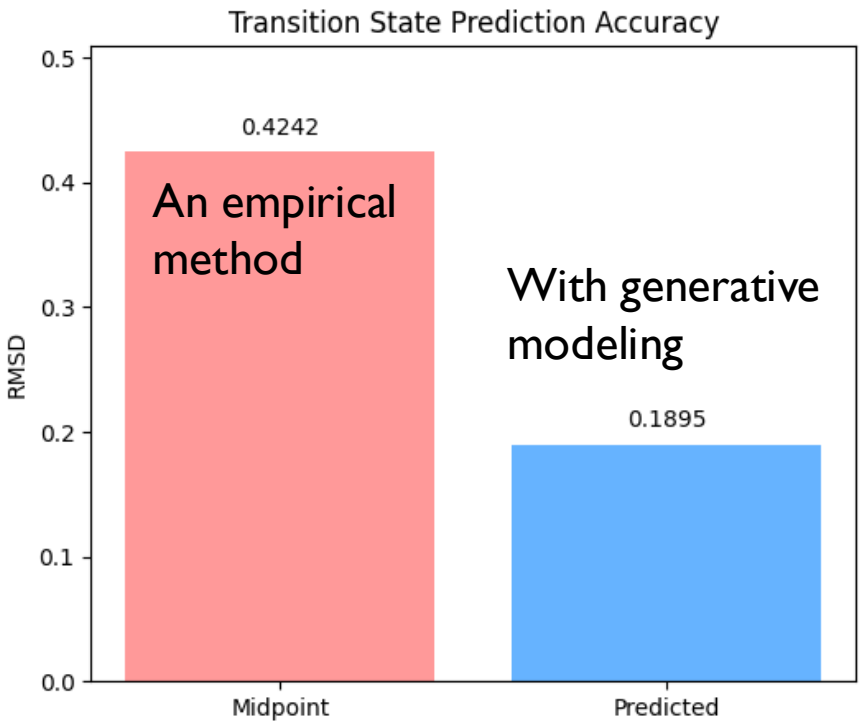


Should be easily loadable on the GPU with a GNN

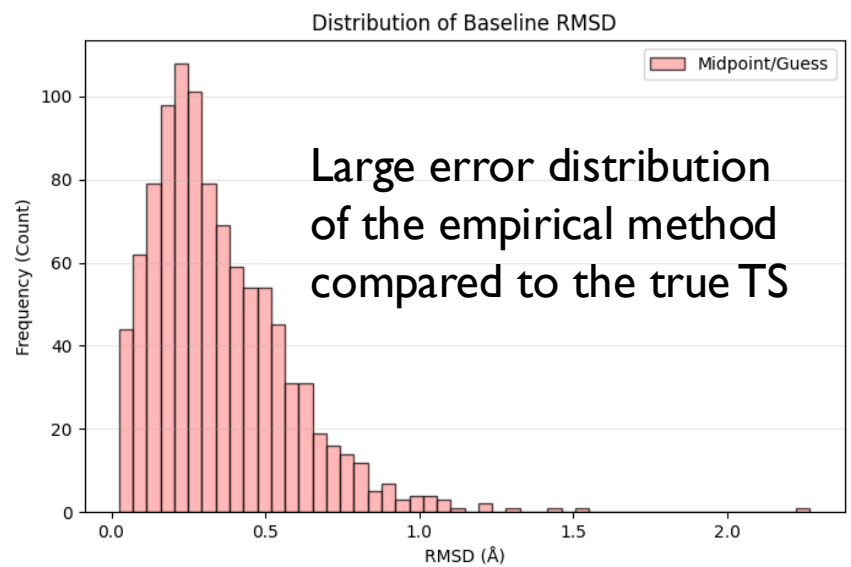
Predicting transition states is hard

Guess the TS and learn a model to move to the true TS

$$\mathrm{RMSD}(\mathbf{v}, \mathbf{w}) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|v_i - w_i\|^2}$$
$$= \sqrt{\frac{1}{n}\sum_{i=1}^{n}((v_{ix} - w_{ix})^2 + (v_{iy} - w_{iy})^2 + (v_{iz} - w_{iz})^2)}$$

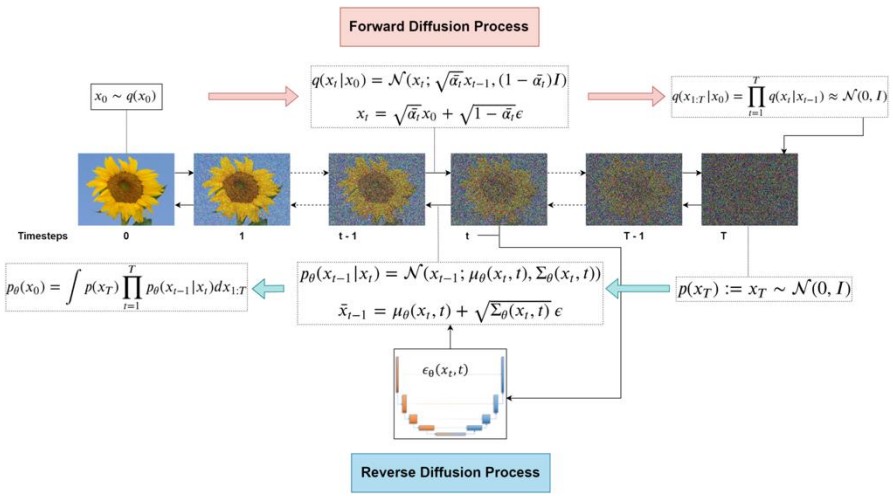


A baseline to beat



There are papers that report RMSD in the order of 0.01!

Generative approaches



- Start from noise, gradually denoise to recover the TS structure
- Many denoising steps
- Has been used with molecular data

- Learn a continuous vector field that transports one structure into another
- Start from an initial guess x_0
- Learn a velocity field $u(x, t)$
- Integrate an ODE to reach the TS

Encoding molecular data

Equivariant GNN (EGNN)

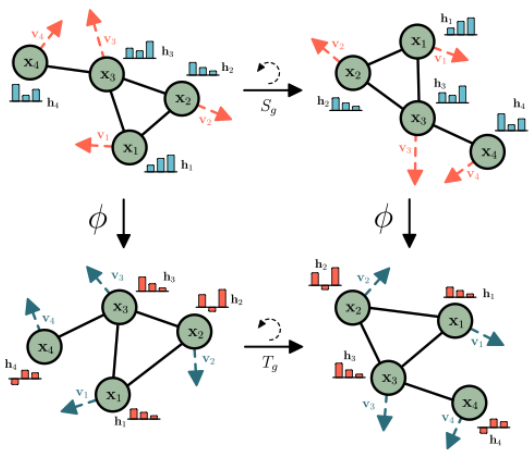


Figure 1. Example of rotation equivariance on a graph with a graph neural network ϕ

Why equivariance matters:

- Physics does not depend on coordinate frame
- The same molecule rotated should give the same prediction (up to rotation)

Neural networks that respect physical symmetries:

- Translation
- Rotation
- Permutation of atoms

$$f(Rx + t) = R\,f(x)$$

- Relative geometry (translation)

$$r_{ij} = x_i - x_j, \qquad d_{ij} = \|x_i - x_j\|^2$$

- Message passing

$$m_{ij} = \phi_m(h_i, h_j, d_{ij})$$

- Node feature update (perm inv)

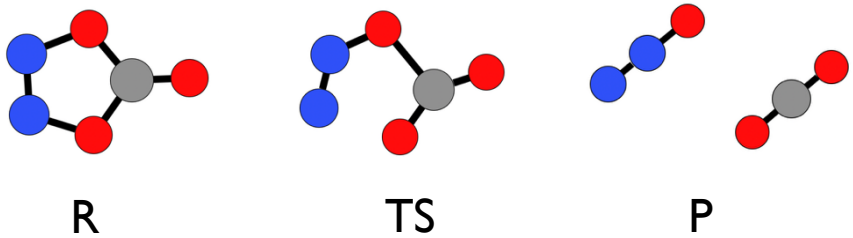
$$h_i^{(l+1)} = h_i^{(l)} + \phi_h\left(\sum_j m_{ij}\right)$$

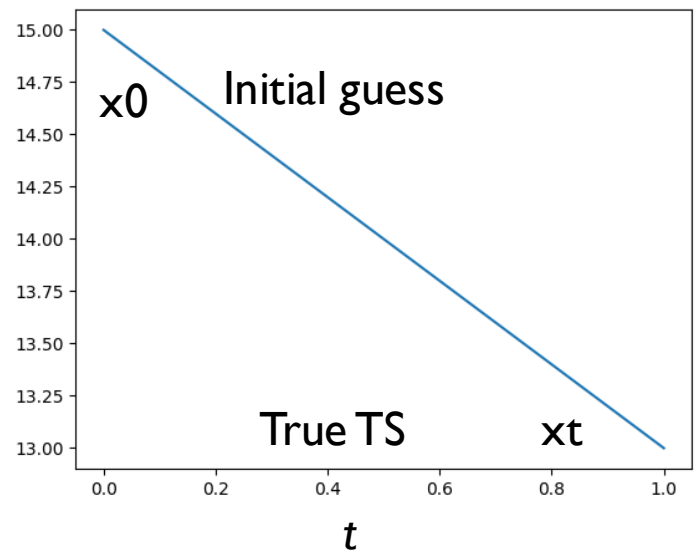
- Equivariant coordinate update

$$x_i^{(l+1)} = x_i^{(l)} + \sum_j (x_i - x_j)\,\phi_x(m_{ij})$$

Flow matching: learning how atoms move

Instead of generating a transition state from scratch, we start from a reasonable guess and learn how atoms should move toward the true transition state.





Interpolation path

$$x_t = (1-t)\,x_0 + t\,x_{TS}, \quad t \in [0,1]$$

Learn the velocity field

$$u_\theta(x_t, t \mid c) \qquad c = \{x_R,\ x_P,\ Z\}$$

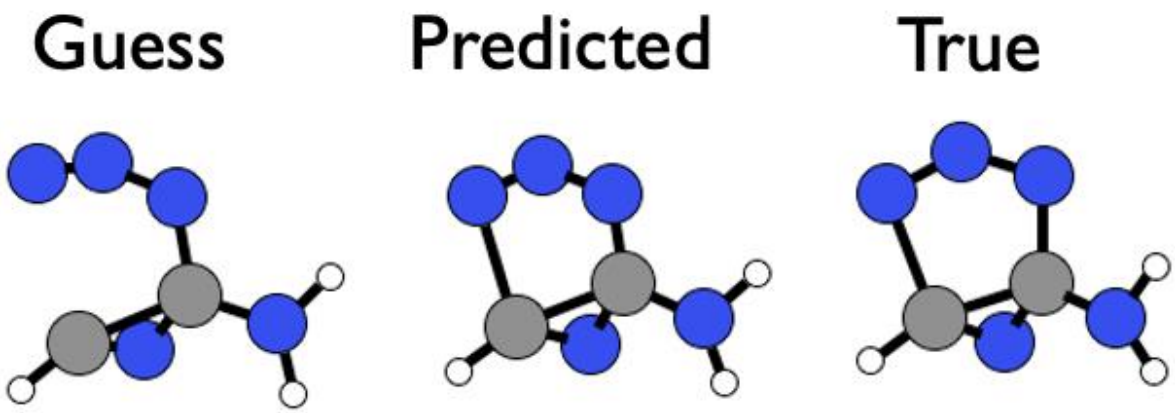
Target velocity field

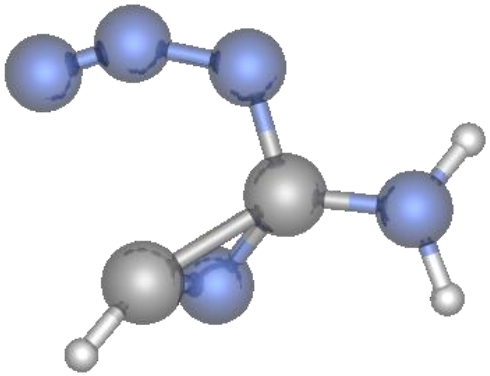
$$u^*(x_t, t) = \frac{dx_t}{dt} = x_{TS} - x_0$$

Loss (randomly sample along the path)

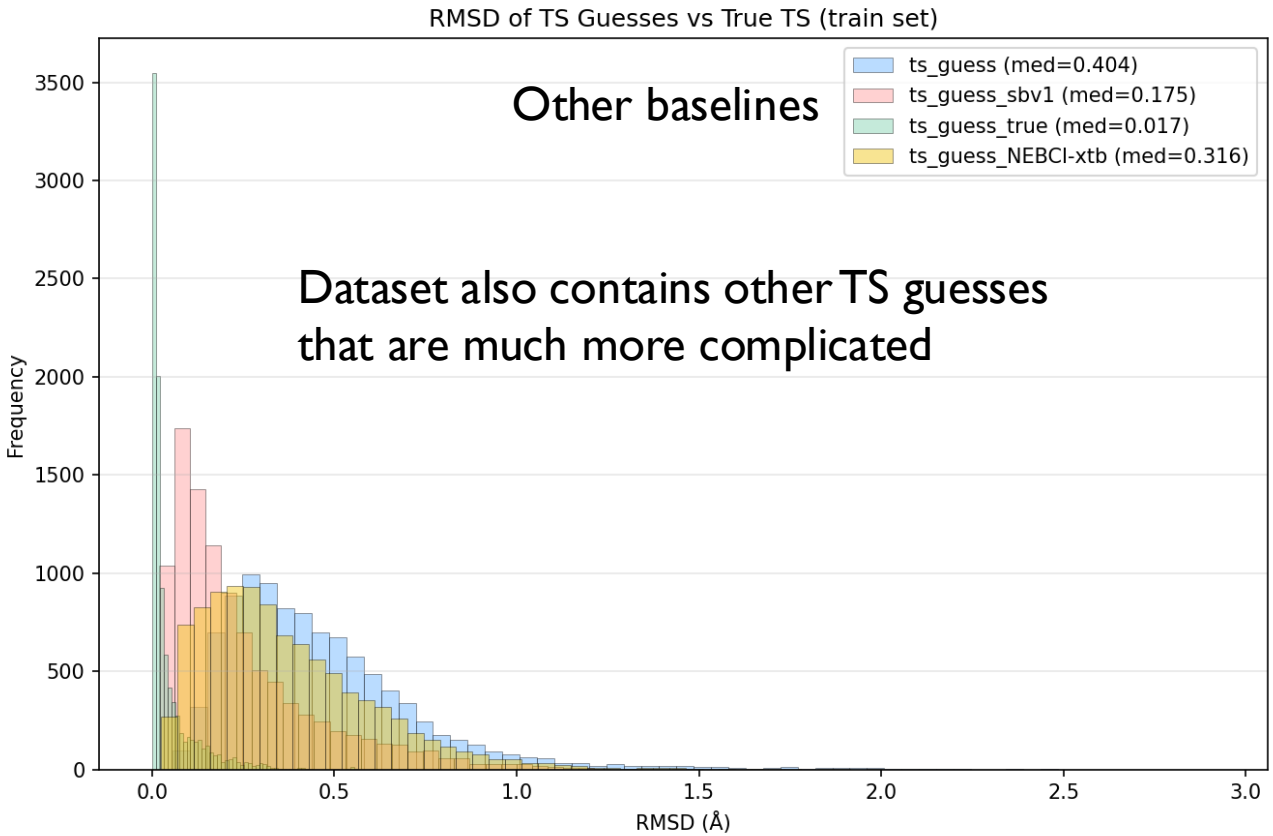
$$\mathcal{L} = \mathbb{E}_{t\sim\mathcal{U}(0,1)}\left[\left\|u_\theta(x_t, t \mid c) - (x_{TS} - x_0)\right\|^2\right]$$

Example of good TS prediction

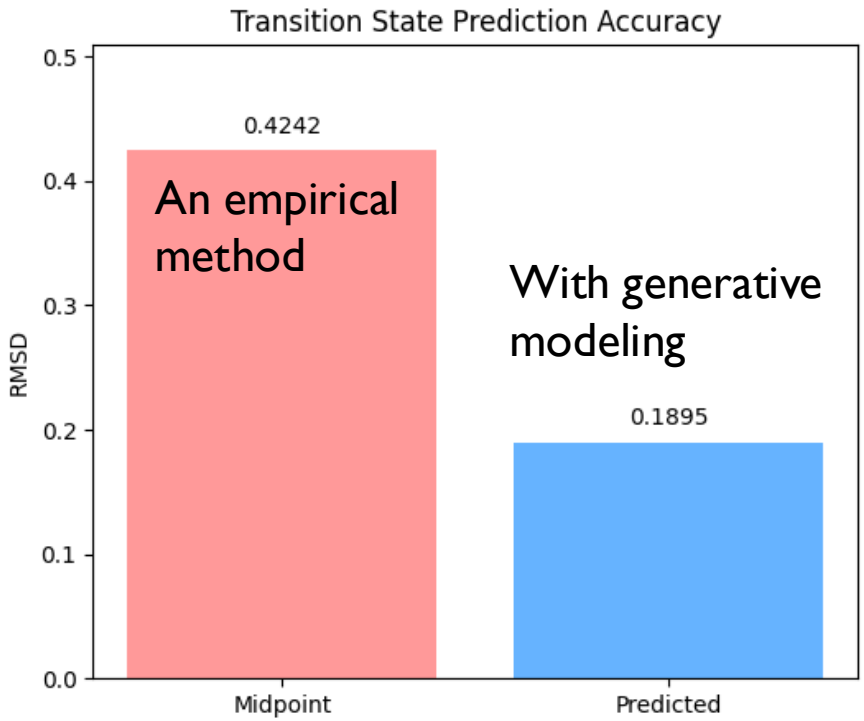




Are there better guesses?



What are other things to work on?



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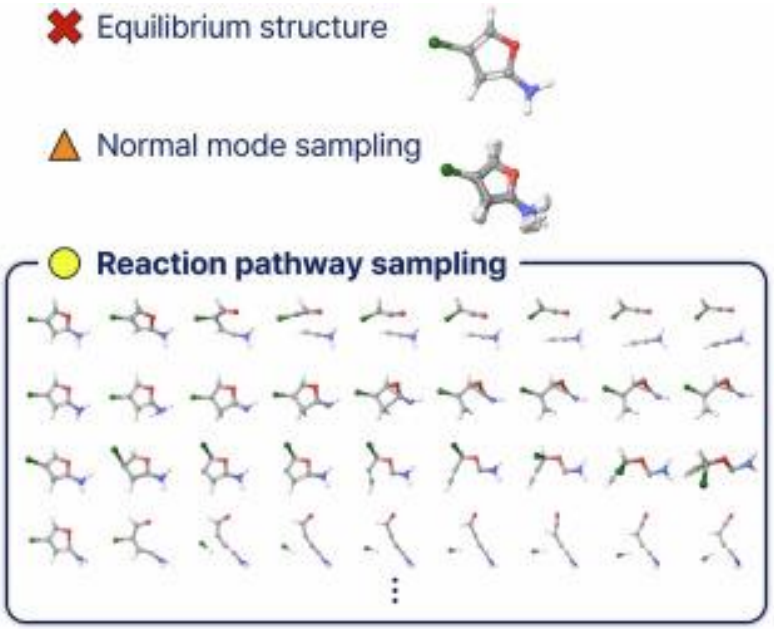
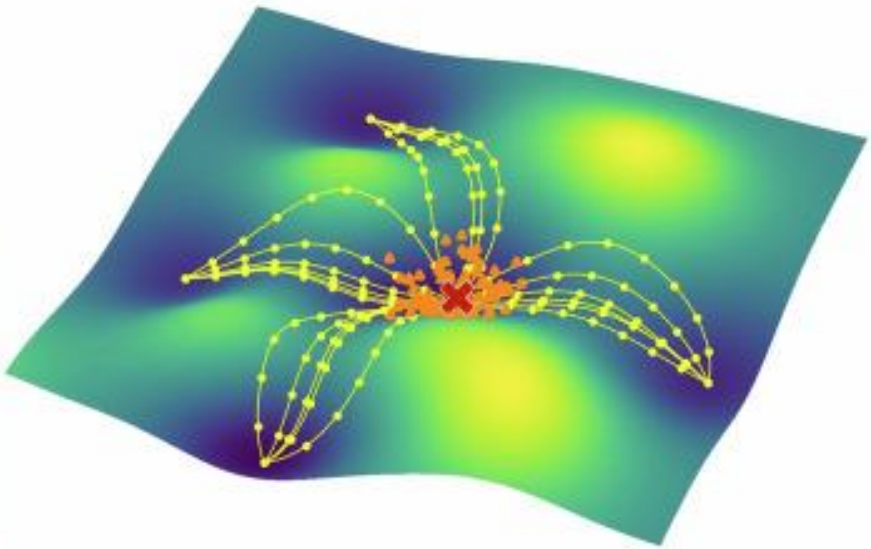
- How to increase transferability?
  - Inclusion of padding in your representation
- How do you encode your connection in your EGNN?
  - fully connected can be expensive and reduce locality for TS prediction
  - Maybe use kNN – other approaches?
- What is a good initial guess?
- Is the linear interpolation a good guess for flow matching?
- Is flow matching the best approach at all?
- Any improvements to EGNN?

Plan to use the Halo8 dataset

Now data with over 20,000 reactions (20 million points)





Predicting the whole reaction pathway

$x(t), \quad t \in [0,1]$

For each reaction: frames are $\{x^{(k)}\}_{k=0}^{K}$

$t_k = \frac{k}{K}$

Supervised samples of the form

$(x(t_k),\ t_k,\ c) \quad c = \{x_R,\ x_P,\ Z\}$

Define conditional velocity field

$u_\theta(x, t \mid c)$

Target velocity field from pathway data

$u^{\backslash *}(x(t_k), t_k) \approx \frac{x^{(k+1)} - x^{(k)}}{t_{k+1} - t_k}$ Finite differences?

Loss using target velocity field

$$\mathcal{L} = \mathbb{E}_{\text{rxn}} \mathbb{E}_k \left[\left\| u_\theta(x^{(k)}, t_k \mid c) - \frac{x^{(k+1)} - x^{(k)}}{\Delta t} \right\|^2 \right]$$

Include other conditions like energy, forces

Inference:

Start from reactant geom

$x(0) = x_R$

Integrate learned ODE

$$\frac{dx}{dt} = u_\theta(x, t \mid c)$$

Sample frames along traj

$x(t_1), x(t_2), \ldots, x(1)$

- Won't be using the linear bridge approximation

Timeline?